Beta test plan

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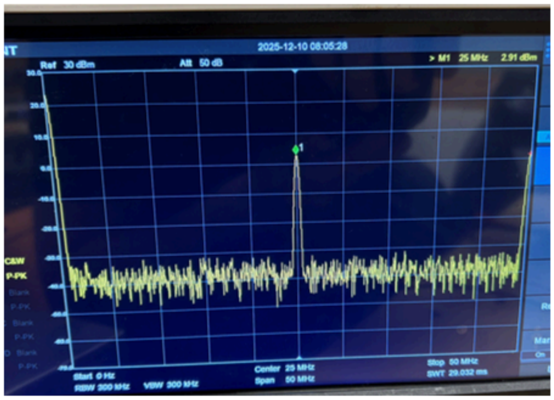
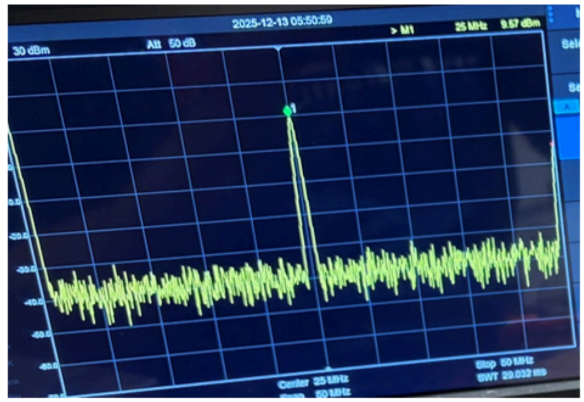
1. Results of alpha testing

a. RF transmitter

In the process of testing, we determined that we need to increase the supply voltage of the first stage PA to allow it to produce enough power for the second stage to amplify it to the final 30 dBm. Initially, we tested 5V - 9V:



At Vdd = 5V, the Pout is around 3 dBm



Highest Pout was at Vdd = 9V, reaching 9.5 dBm

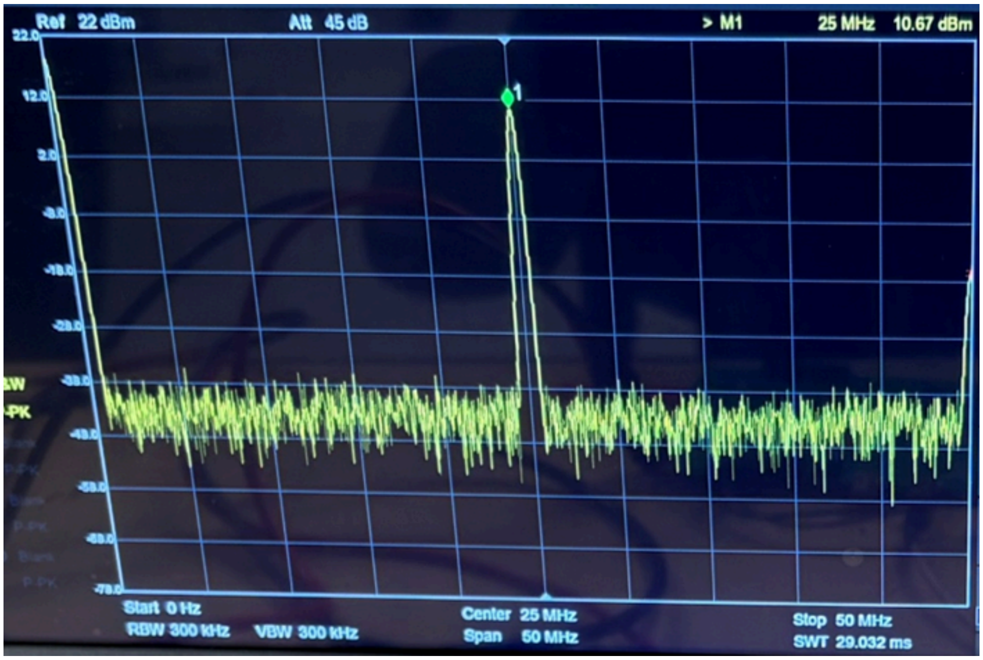
Iq = 51 mA

DDS and PA1 Tx current at Vdd = 5V

DDS and PA1 Tx current at Vdd = 9V

After increasing the voltage even higher to 13V, we achieved the output power of 10.67 dBm:

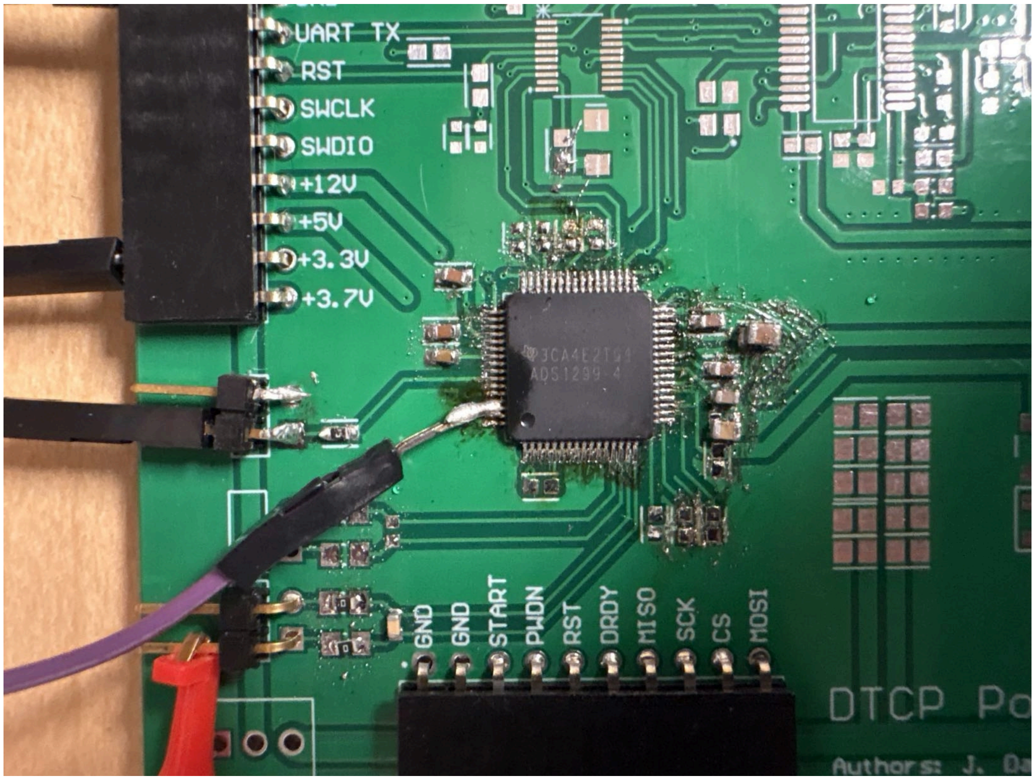
Then, the output spectrum of the signal coming out of the second stage PA is shown below (a 20dB attenuator was used to protect the spectrum analyzer):



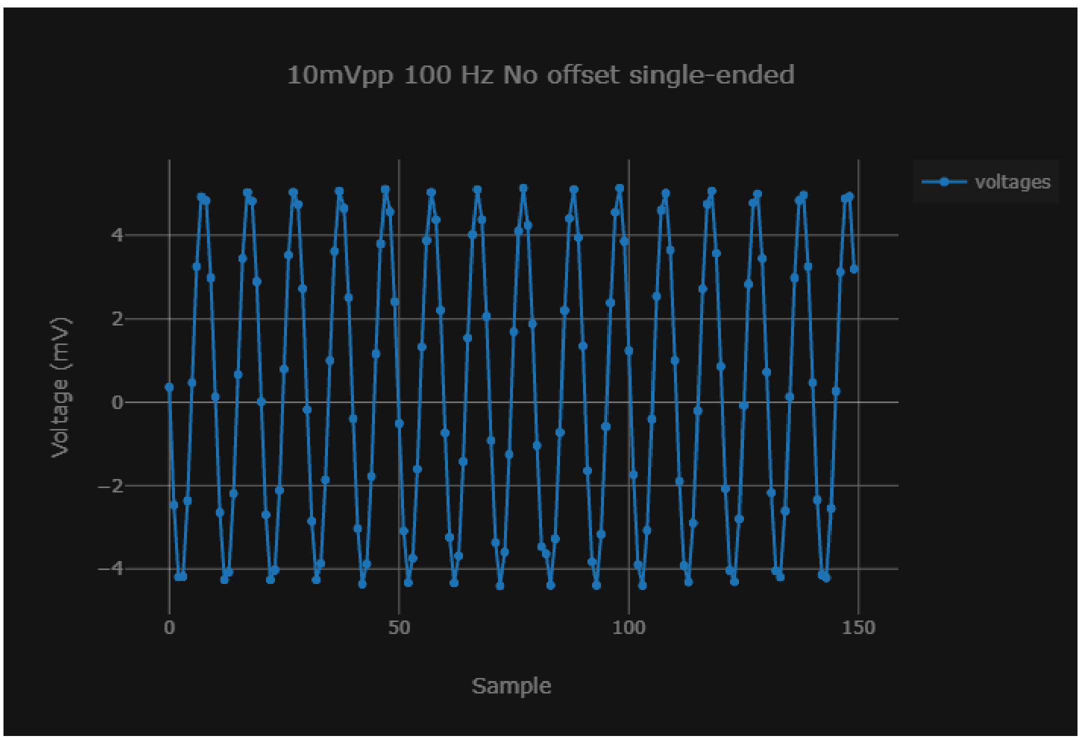
As we can see, the output power is 30.67 dBm (after adding the 20 dB absorbed by the attenuator).
During transmission, the current consumption is 100 mA for the 3.3V supply and 330 mA for the 13V supply:

When the device is not transmitting power, the idle current is below 10 mA thanks to the use of power switches that disable the DDS and power amplifiers.

b. EMG recording system
In the process of testing, we realized that the AFE chips needs +2.5V DC bias applied to the inputs to be able to record the signals. We had to manually solder a wire to the AFE bias pins and then connect it to the negative analog input:



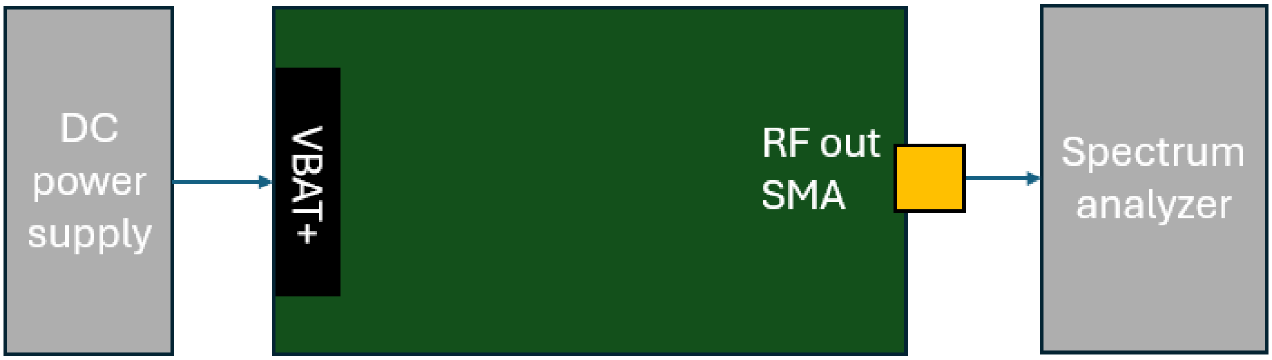
After doing that, we were able to record signals mimicking an EMG signal with high accuracy:



2. RF transmitter circuit validation

a. Expected behavior
After providing +3.7V DC to the battery terminals, the device powers up and starts transmitting a 25 MHz RF signal at >30 dBm.

b. Test procedure
Set up the devices as follows:



The goal is to only provide DC power emulating a battery, let the device handle all internal signals and see it output the target signal (25 MHz at 30+ dBm).

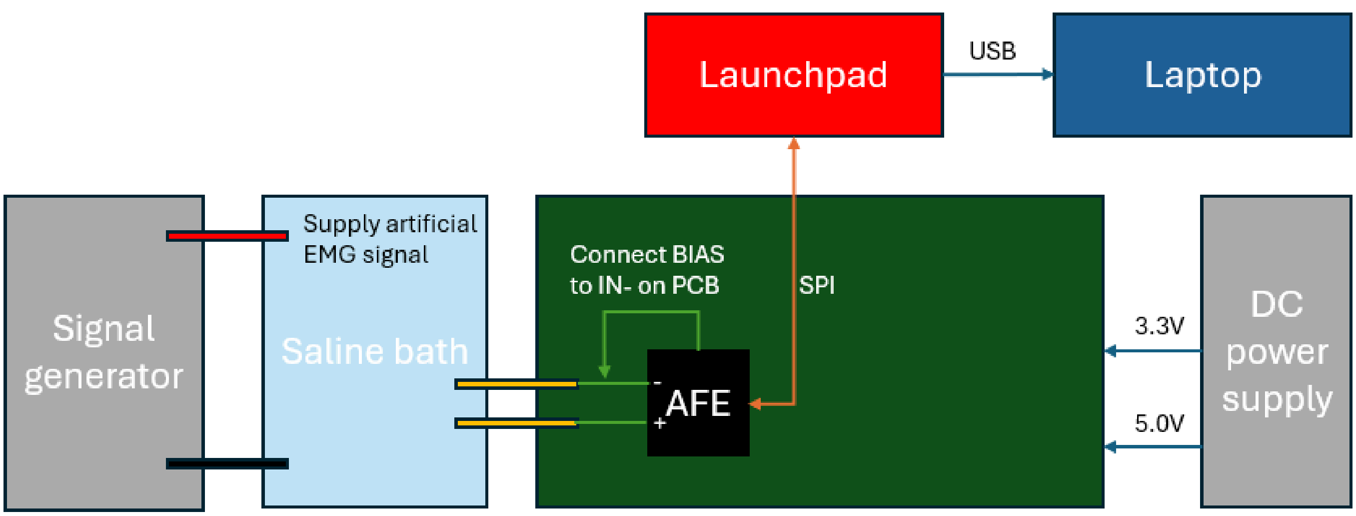
3. EMG recording validation

a. Expected behavior

The AFE should be able to read an AC signal in saline medium with an SNR of at least 10 dB (e.g noise floor of 0.5 mV for 5mV EMG signals) to emulate recording a real muscle.

b. Test procedure

Use a power supply to supply 3.3V and 5V DC for the digital and analog parts of the board. Use a function generator to supply a 100 Hz, 10 mVpp sine wave to the input of a channel through the saline medium. Debug the firmware project using CCS and use the Graph functionality to continuously plot the recorded values.



The goal is to obtain a graph of an artificial EMG signal on the laptop connected to the AFE via a Launchpad.